Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 09-03-2026

REAL TIME EVENT SYNCHRONIZATION ENGINE

SDG: 4 Quality Education

Presented By:

M.V SAI AAKASH – VTU26363 – 23UECS1126

Courses Handling Faculty:

Dr.S.HEMAMALINI-M.E,Ph.D.,

INTRODUCTION

1

2

3

4

5

6

7

8

9

In many modern applications, users work simultaneously on shared data, making real-time synchronization a crucial requirement. Traditional web systems often require manual page refreshes to view updates, which leads to delays and inconsistent information. This project addresses the problem by providing a system where event updates are instantly reflected across all connected users. The **Real-Time Event Synchronization Engine** ensures that whenever an event is created, modified, or deleted, the changes are automatically synchronized in real time. This improves collaboration, efficiency, and user experience. Such systems are widely used in applications like collaborative tools, live dashboards, shared calendars, and online platforms. The system is built using technologies like Node.js, Express.js, and MongoDB to manage and synchronize event data effectively. Overall, this project demonstrates how real-time communication can enhance modern web applications.

PROBLEM STATEMENT

1
2
3
4
5
6
7
8
9

- Many existing systems do not provide real-time updates when multiple users work on the same data.
- Users must manually refresh the page to see changes made by others.
- This creates delays and inconsistent information across different users.
- Managing events becomes difficult when updates are not instantly synchronized.
- Lack of real-time communication reduces collaboration efficiency.
- It may also lead to data conflicts when multiple users modify events simultaneously.
- Therefore, there is a need for a system that automatically synchronizes event updates in real time.
- A Real-Time Event Synchronization Engine can solve this problem by ensuring instant updates and better data consistency.

OBJECTIVE(S)

1
2
3
4
5
6
7
8
9

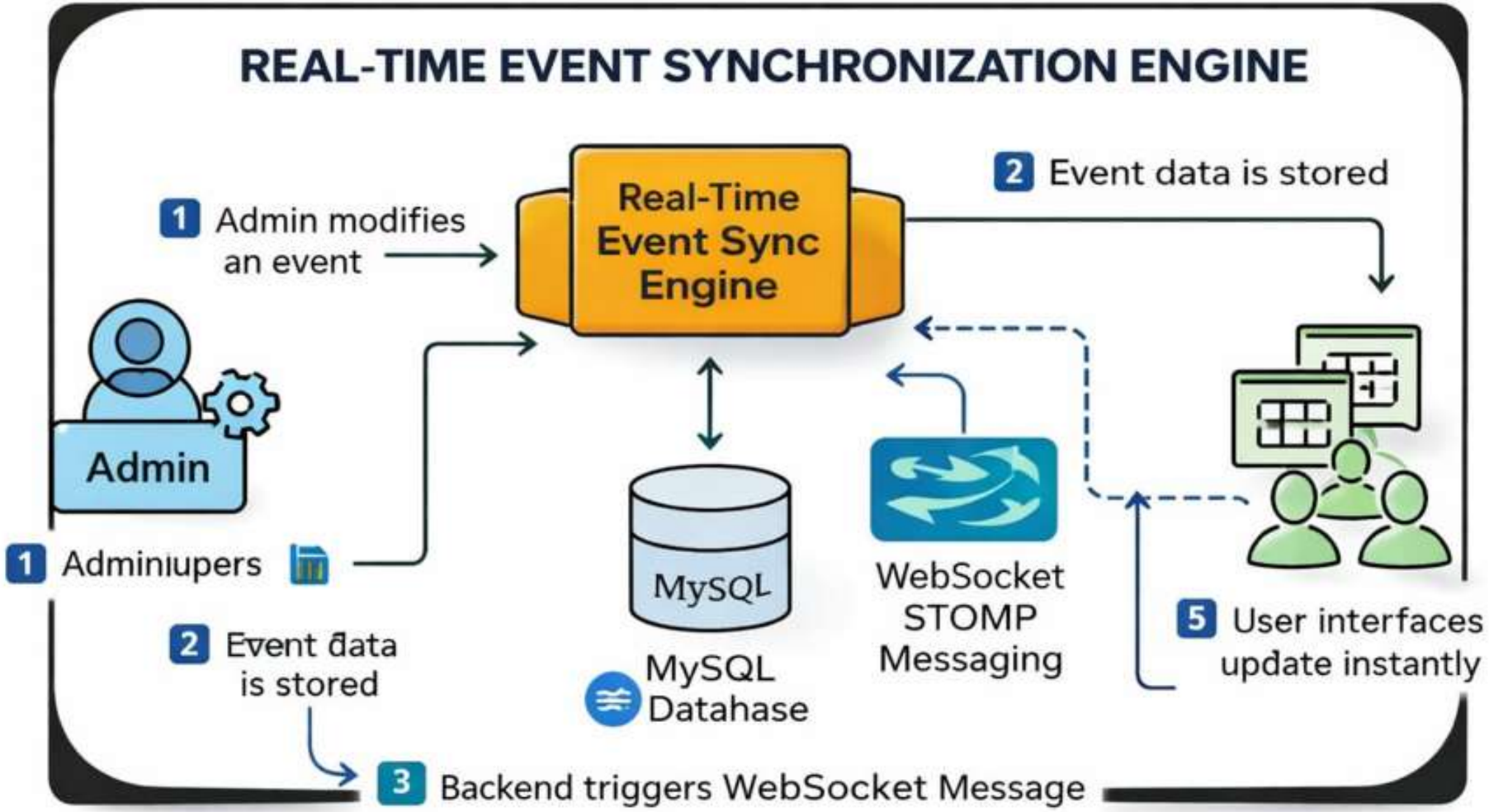
- **To design and develop** a real-time event synchronization system that allows multiple users to create, update, and manage events simultaneously.
- **To implement** instant data synchronization so that all event changes are automatically updated across connected clients without page refresh.

PROPOSED SYSTEM

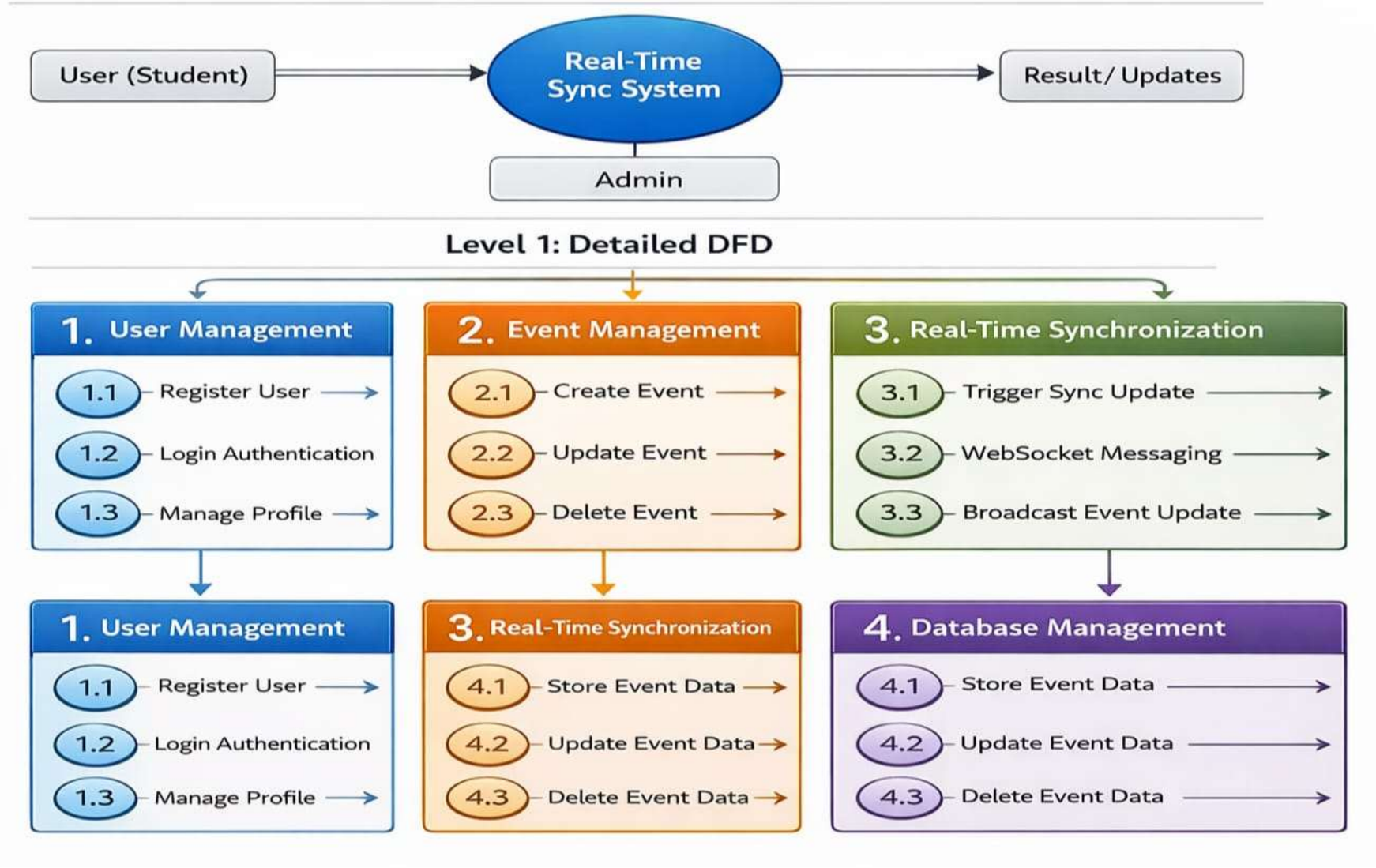
1
2
3
4
5
6
7
8
9

- The system provides a real-time platform where multiple users can create, update, and delete events simultaneously.
- It uses WebSocket-based communication to instantly synchronize data between the server and all connected clients.
- Whenever an event is modified by one user, the update is automatically broadcast to all other users in real time.
- The system stores event data securely in a database to maintain consistency and reliability.
- This solution eliminates the need for manual page refresh to view updates.
- It improves collaboration by ensuring that all users always see the latest event information.
- The system reduces data conflicts that occur when multiple users work on the same data.
- Compared to traditional systems, it provides faster updates, better efficiency, and an improved user experience.

PROPOSED SYSTEM ARCHITECTURE



WORKFLOW


User (Student)
Real-Time Sync System
Result/ Updates
Admin
Level 1: Detailed DFD
1. User Management
1.1 Register User
1.2 Login Authentication
1.3 Manage Profile
2. Event Management
2.1 Create Event
2.2 Update Event
2.3 Delete Event
3. Real-Time Synchronization
3.1 Trigger Sync Update
3.2 WebSocket Messaging
3.3 Broadcast Event Update
1. User Management
1.1 Register User
1.2 Login Authentication
1.3 Manage Profile
3. Real-Time Synchronization
4.1 Store Event Data
4.2 Update Event Data
4.3 Delete Event Data
4. Database Management
4.1 Store Event Data
4.2 Update Event Data
4.3 Delete Event Data

TECHNOLOGIES USED

1
2
3
4
5
6
7
8
9

Frontend:

1 . HTML5
2 . CSS3
3 . JavaScript

Backend:

1 . PHP
2 . MySQL
3 . Apache

Tools:

1 . XAMPP Server
2 . Visual Studio Code

Software Requirements:

1 . Windows
2 . Web Browser

MODULE DESCRIPTION

Module 1: User Management
Handles user registration, login, and authentication.

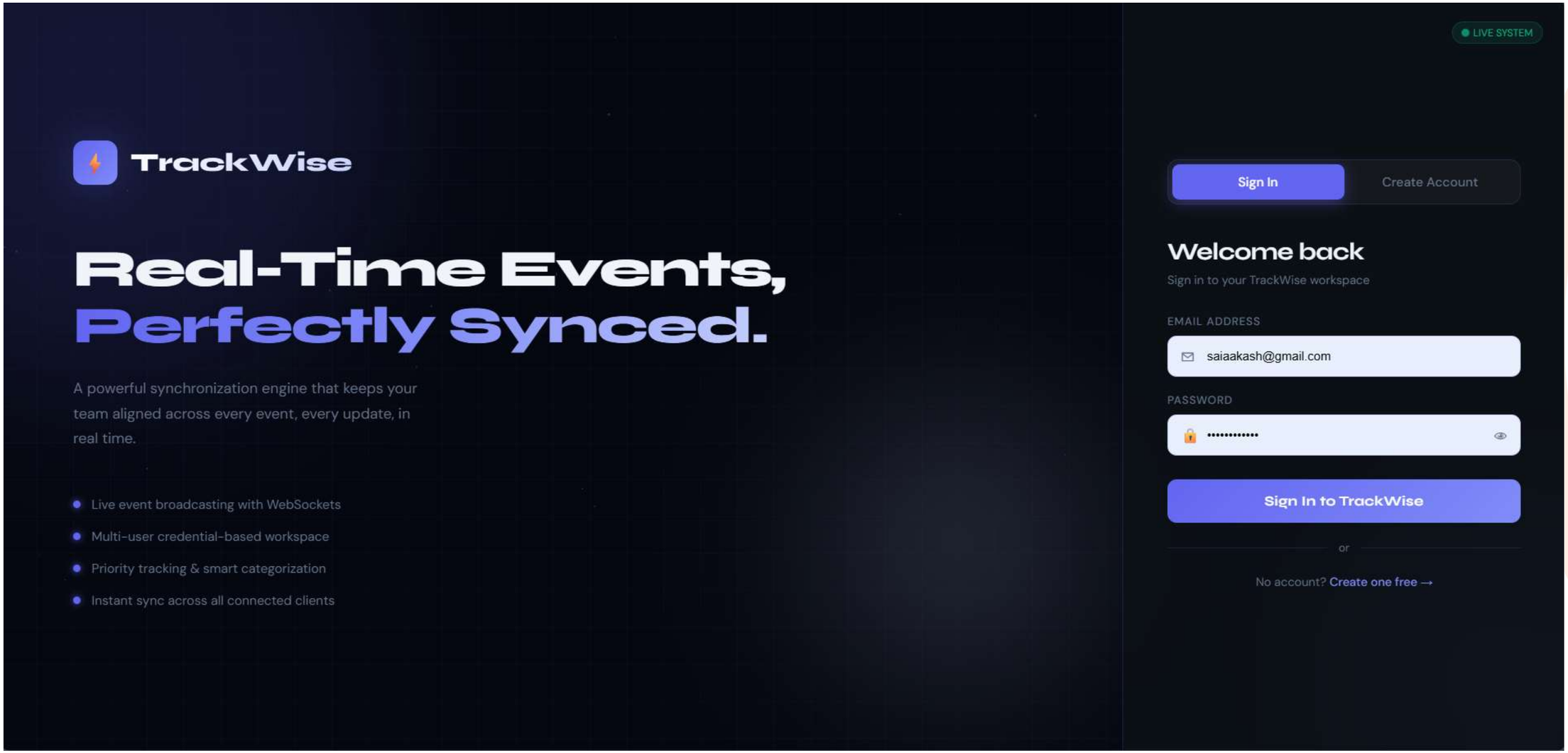
Module 2: Core Event Management
Allows Admin to create, update, and delete events.

Module 3: Real-Time Synchronization Module
Uses WebSocket to send event updates instantly to all users.

Module 4: Database Management
Stores event details and user information securely.

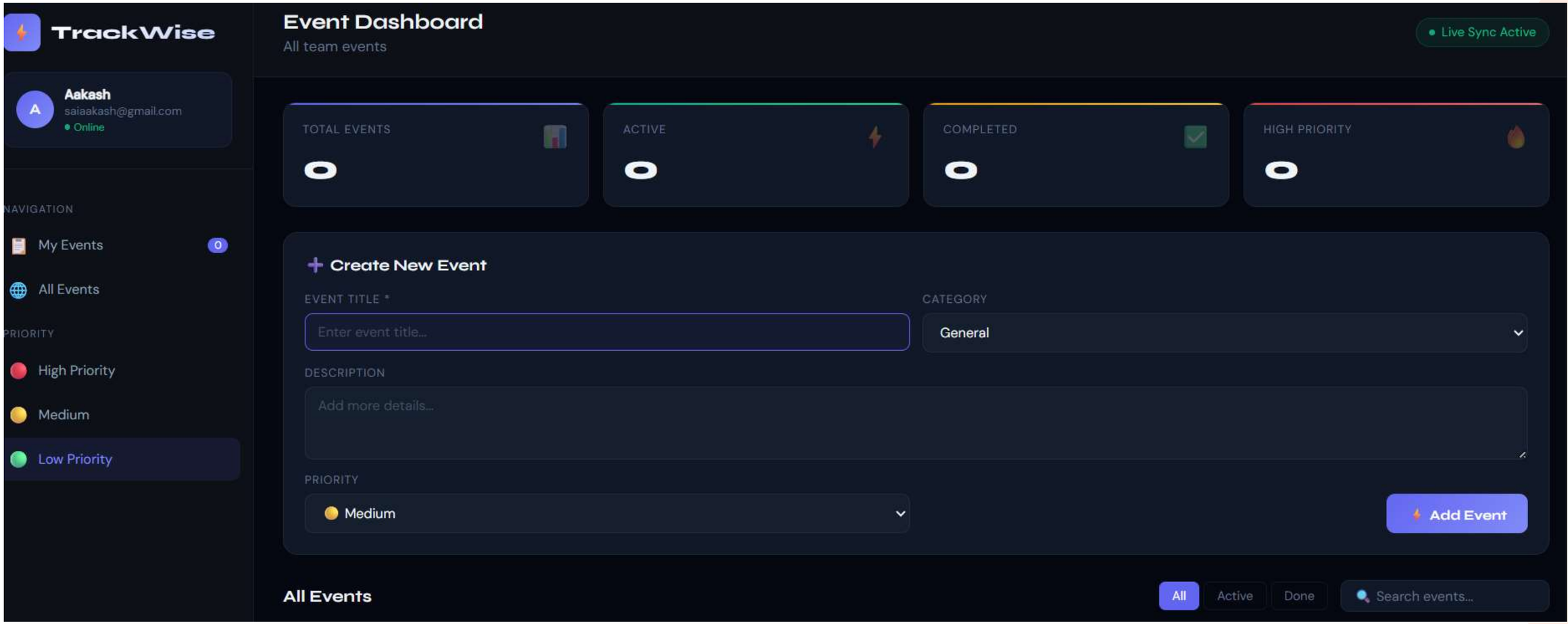
IMPLEMENTATION SCREENSHOTS

1
2
3
4
5
6
7
8
9



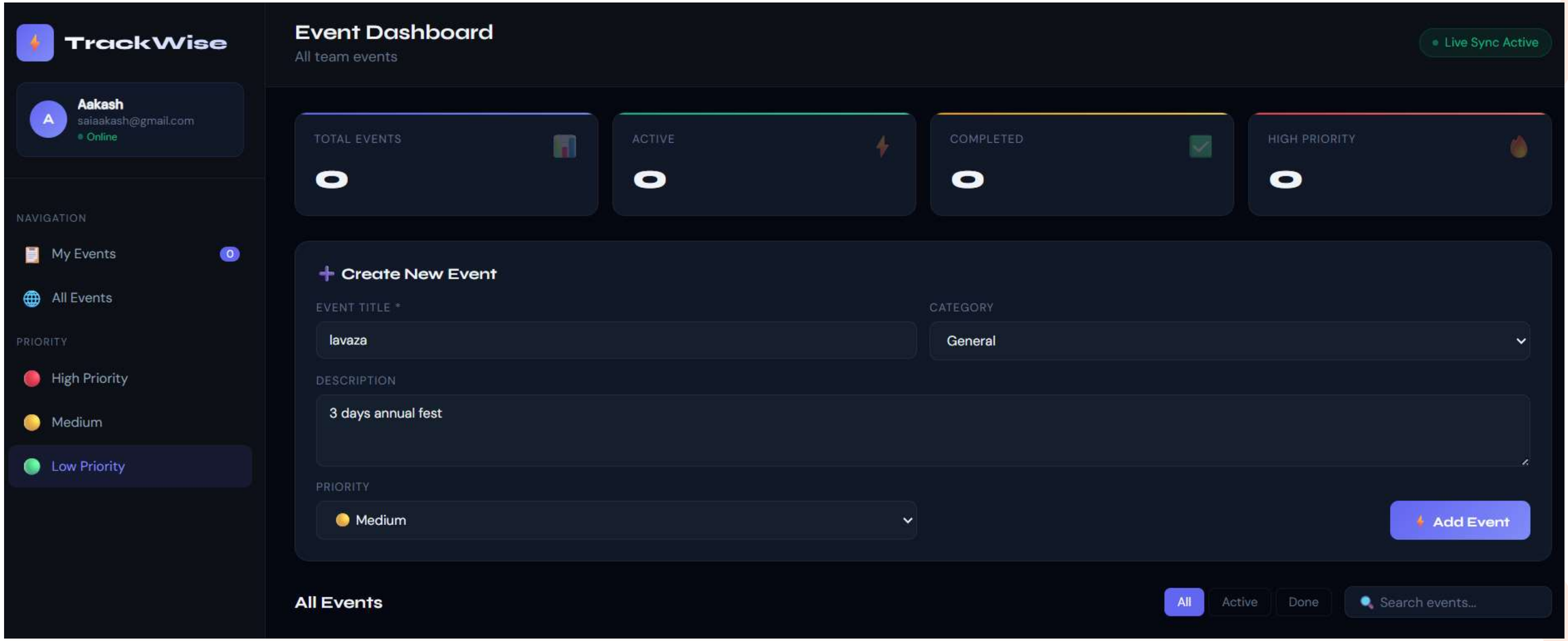
IMPLEMENTATION SCREENSHOTS

1
2
3
4
5
6
7
8(A)
9



IMPLEMENTATION SCREENSHOTS

1
2
3
4
5
6
7
8(B)
9



ADVANTAGES & LIMITATIONS

1
2
3
4
5
6
7
8
9

Advantages:

1 . Instant Event Updates.
2 . No Manual Page Refresh.
3 . Prevents Scheduling Conflicts.
4 . Improved User Experience.
5 . Scalable System.

Limitations:

1 . Requires Stable Internet Connection.
2 . WebSocket Configuration Complexity.
3 . Security Challenges in Real-Time Systems.
4 . Server Load Handling Required.

Thank you